

NYC Mobility Pipeline & Insights

Taxi Zone Prioritisation for Targeted Mobility Interventions

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Business Question

Which NYC taxi zones should be prioritised for targeted mobility interventions, based on taxi demand volume, time-of-day reliance, trip distance patterns, and comparison with subway, bus and Citi Bike activity?

Audience: NYC mobility/transport planning department **Decision supported:** shortlisting zones for further service-planning review (not final infrastructure decisions)

Summary

This project builds a reproducible data pipeline to analyse NYC mobility data and identify which taxi zones should be prioritised for targeted intervention. Yellow and green taxi trip records were used as the anchor datasets, supported by taxi zone lookup data, subway hourly ridership, bus hourly ridership and Citi Bike trip data — over **16.8M+ combined records**.

The pipeline sourced, audited, cleaned and transformed raw public datasets into engineered summary tables for analysis, then classified taxi zones into four intervention categories based on percentile-based demand signals (high volume, late-night reliance, short-trip concentration, long-trip/connectivity needs).

Key Findings

Taxi demand is **highly concentrated in Manhattan** (3.2M of 3.7M total trips), with distinct demand signals varying by zone and time of day

58.6% of Manhattan trips are short-distance (<2 miles) — candidates for bike/bus alternative review

Queens shows the strongest long-trip pattern (79.3%), driven largely by JFK/LaGuardia airport travel

Demand peaks in the **evening (17:00–21:00)**, not just the morning commute

Final output classified **73 candidate zones** into 4 intervention categories: taxi pickup/drop-off management (29 zones), late-night public transport support (27 zones), short-trip alternative review (7 zones), and public transport connectivity review (10 zones)

Pipeline & Methods

Sourcing: Direct downloads + Socrata API pagination for large MTA datasets; Google Drive-based reproducible file structure

Loading: DuckDB for efficient large Parquet file querying; pandas for smaller datasets

Auditing: Percentage-based severity classification (None/Low/Medium/High) across 6 datasets

Cleaning & feature engineering: Standardised taxi records, engineered time-period and trip-distance features, joined zone lookup for geographic enrichment

Analysis: 4-layer SQL analysis (demand patterns → supporting mode summaries → mode comparisons → intervention priority classification) across 16 engineered DuckDB tables

Governance: Explicit data minimisation, privacy-aware aggregation, and responsible-wording controls to avoid overclaiming from aggregate data

Tools: Python, SQL, DuckDB, pandas, GeoPandas, Google Colab

Repository Contents

`NYC_Mobility_Pipeline.ipynb` — full analysis notebook (data pipeline, SQL, visualisations)

`NYC_Mobility_Pipeline_and_Insights.pdf` — full written report with all figures, tables and appendices

Data Sources

NYC TLC Yellow & Green Taxi Trip Records · MTA Subway Hourly Ridership · MTA Bus Hourly Ridership · Citi Bike Trip Data · NYC Taxi Zone Lookup

This project used publicly available data and applied privacy-aware aggregation throughout — no individual trip paths, passenger identities, or driver behaviour were analysed or visualised.